Nome N.º Aluno

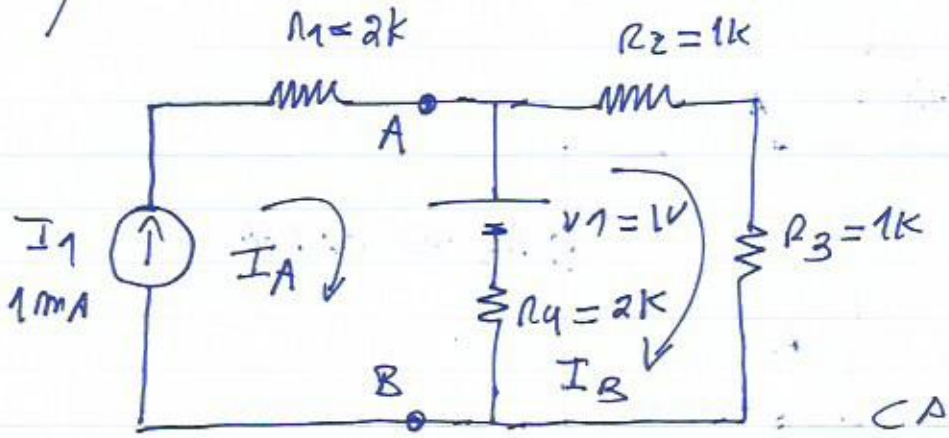
Curso Ano Letivo Data da Avaliação

Prova Escrita de: N.º Folhas

Época:

14)

a)




CALCULO Vth = ?

$$\begin{cases} V_{AB} = V_1 + R_4 \times (I_A - I_B) \\ I_A = I_1 \\ -V_1 + (R_2 + R_3) I_B + R_4 \times (I_B - I_A) = 0 \end{cases}$$

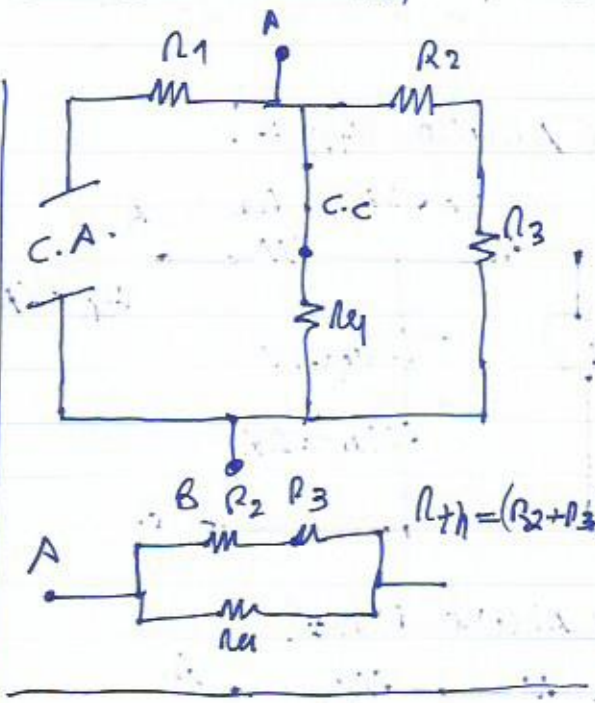
$$I_B \times (R_2 + R_3 + R_4) = V_1 + R_4 I_A$$

$$I_B = \frac{V_1 + R_4 \times I_A}{R_2 + R_3 + R_4} \Rightarrow V_{AB} = V_1 + R_4 \times \left( I_A - \frac{V_1 + R_4 \times I_A}{R_2 + R_3 + R_4} \right)$$

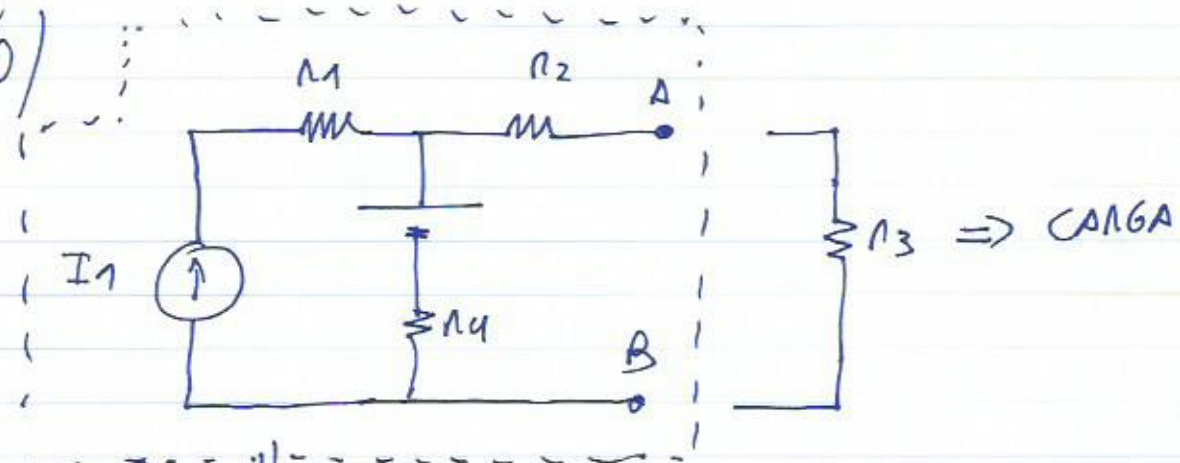
$V_{th} = 1.5\,V$

$R_{th} = 1k$

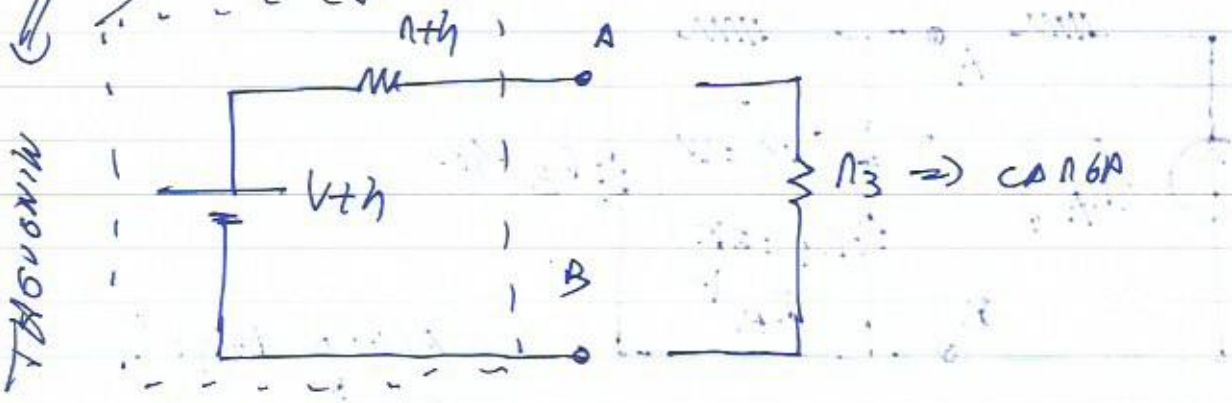
CALCULO Rth: ?



b)

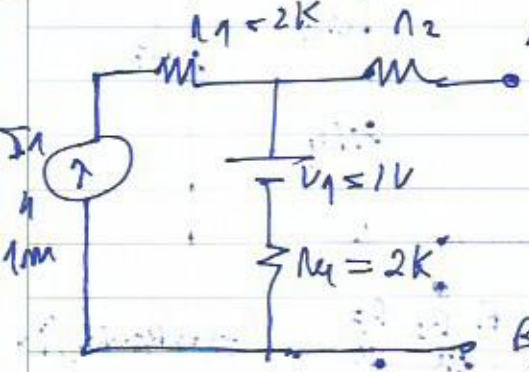



SIMPLIFICADA




TENSIONES DE THEVENIN

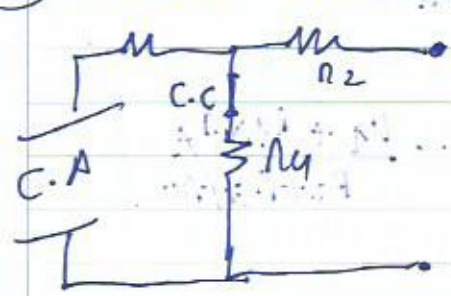
1°) Calculo $V_{th}$ = ?




$V_{th} = V_{AB} = V_1 + R_4 \cdot I_1$

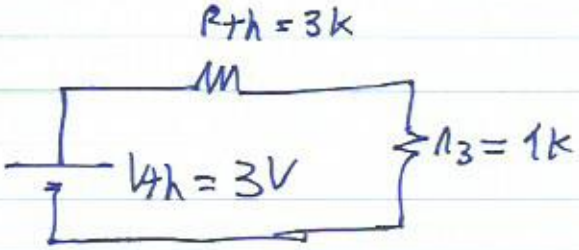
$= 1 + 2k \times 1m = 3V$

2°) Calculo $R_{th}$ = ?




$R_{th} = R_4 + R_2 = 2K + 1K = 3K$

3°) Calculo $I_{R3}$




$I_{R3} = \frac{3}{4k} = 0{,}75mA$

Nome ____ N.º Aluno ____

Curso ____ Ano Letivo ____ Data da Avaliação ____

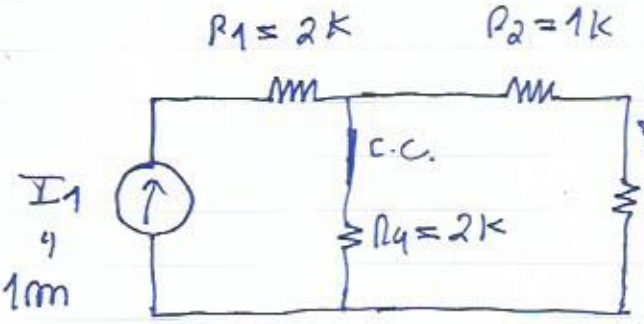
Prova Escrita de: ____ N.º Folhas ____

Época: ____

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C) TEOREMA DA SOBREPOSIÇÃO.

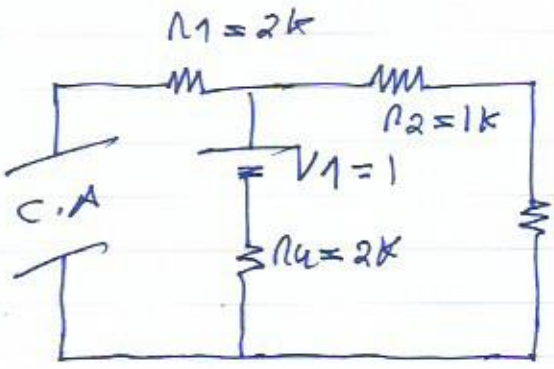
① CONTRIBUTO $I_1$




$$I_{R3}' = \frac{I_1 \times R_4}{R_4 + (R_2 + R_3)} = \frac{I_1 \times 2k}{2k + (1k + 1k)}$$

$$(\downarrow) I_{R3}' = I_1 \times 0.5 = 0.5\,mA$$

② CONTRIBUTO $V_1$




$$I_{R3}'' = \frac{V_1}{R_2 + R_3 + R_4} = \frac{1}{1k + 1k + 2k}$$

$$I_{R3}'' = \frac{1}{4k} = 0.25\,mA$$

$$I_{R3}(\downarrow) = I_{R3}'(\downarrow) + I_{R3}'' = 0.5m + 0.25m = 0.75mA$$